

LECTURE 02 SDLC / Dev. Methodologies

Software Development Life Cycle

Recap



What is Programming Design ?

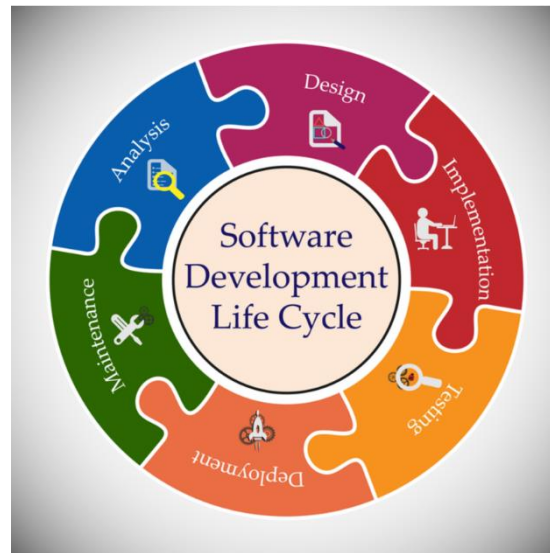
Introduction to Algorithm

Step in Problem Solving

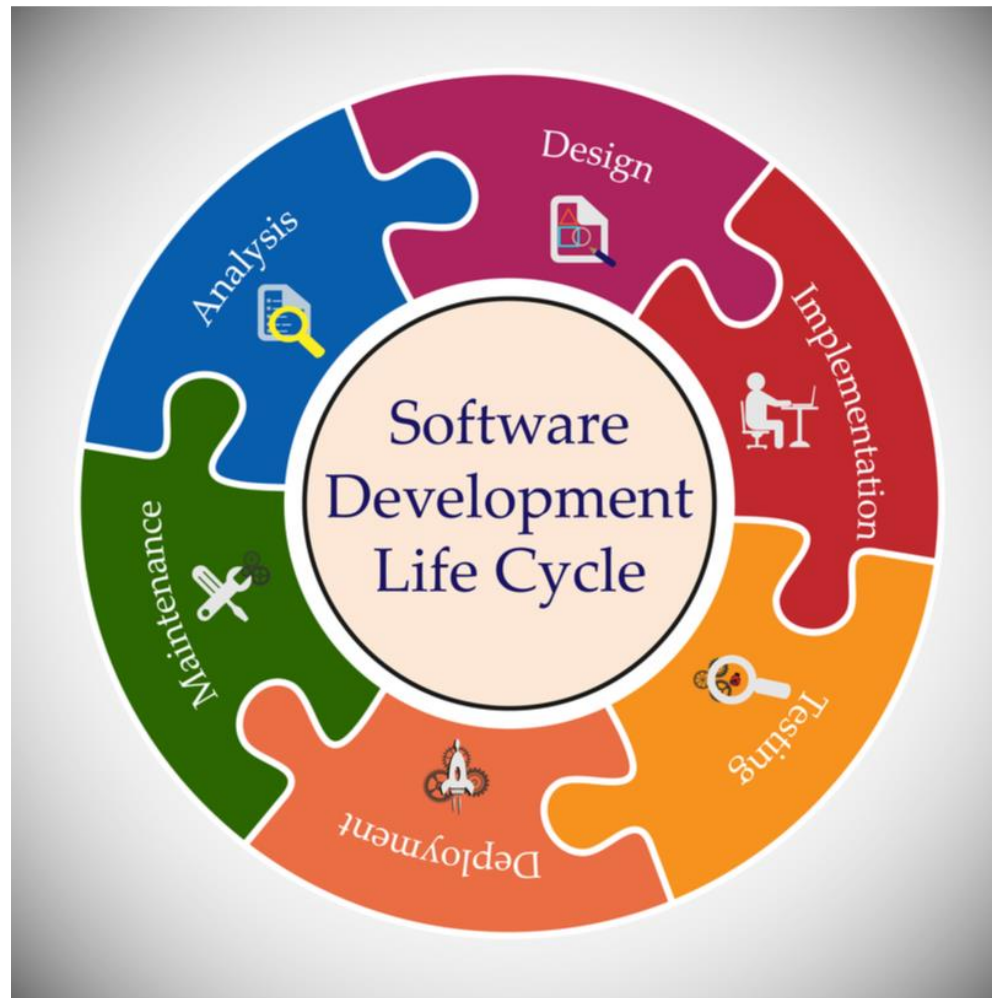
- Define the Problem
- Outline
- Development the outline in the algorithm
- Test the algorithm

What is Software Development Life Cycle

Software Development Life Cycle (SDLC) is a **systematic process or set of phases** that guide the development of software applications from conception to deployment and maintenance.

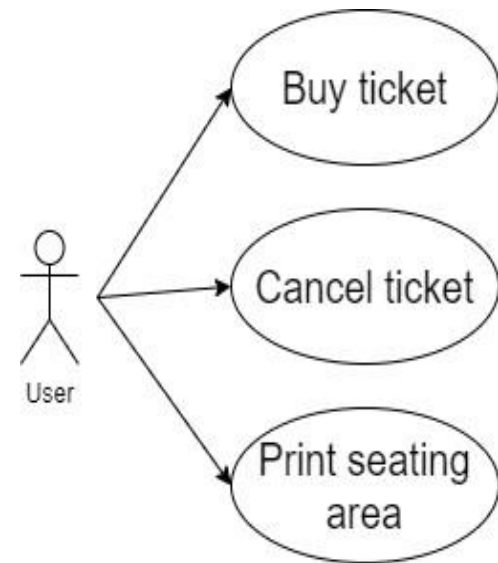


Phases in SDLC



1. Requirements Gathering and Analysis

- Identify and understand the needs and requirements of the software.
- Define the scope of the project and document functional and non-functional requirements



Feasibility Study

1. Technical Feasibility
2. Operational Feasibility
3. Economical Feasibility
4. Organizational Feasibility
5. Legal Feasibility



FR / NFR



- Functional Requirement

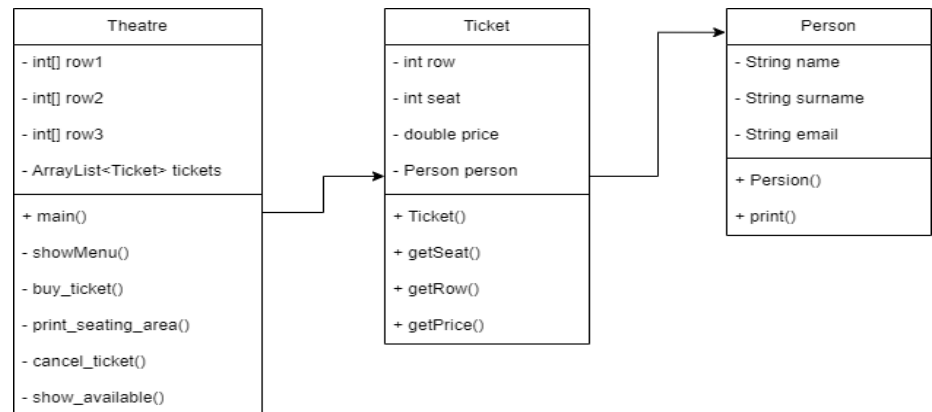
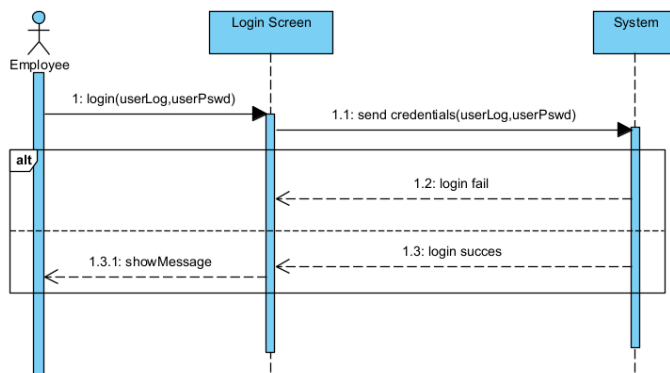
Describe the specific behaviors or functions that a software system must perform

- Non-Functional Requirement

Specify the quality attributes or constraints that the system must satisfy.

2. System Design

- Create a high-level design that outlines the system architecture and components.
- Define data structures, interfaces, algorithms, and overall system behavior.
 - Tools : UML Diagrams – Class Diagram, Sequence Diagram, ER Diagram, DFD, Flow chart



3. Implementation (Coding)

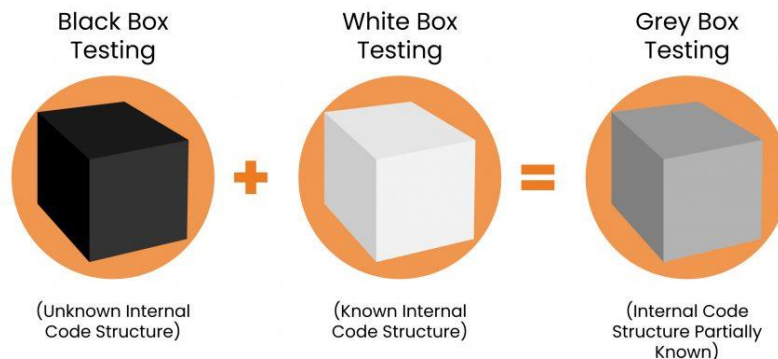
- Write and develop the actual code based on the design specifications.
- Perform unit testing to ensure individual components work as intended.

4. Testing

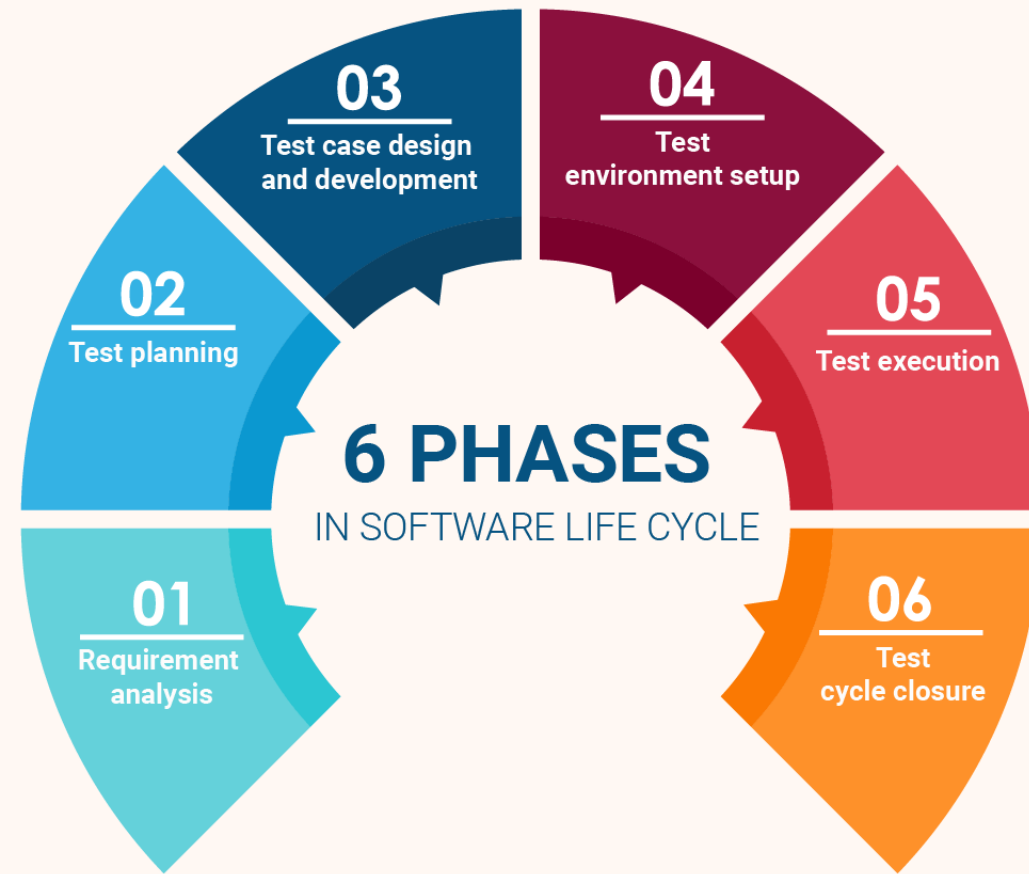
- Conduct various testing phases, such as integration testing, system testing, and user acceptance testing.
- Identify and fix defects or issues found during testing.

White box testing, Black box testing, Grey Box Testing

Types Of Testing Methods



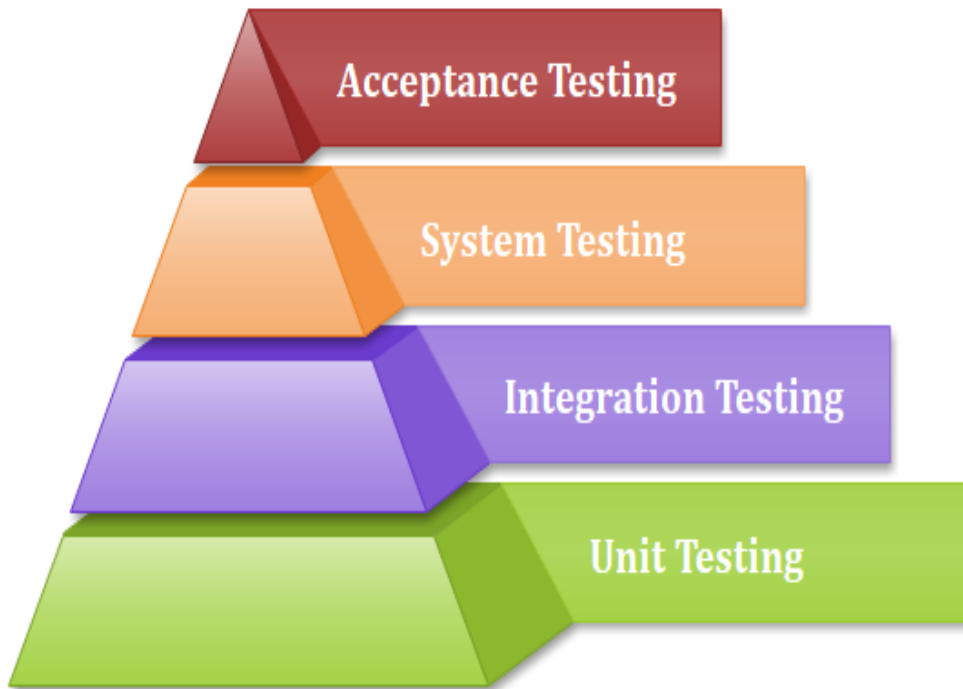
SOFTWARE TESTING LIFE CYCLE



Definition:

The Software Testing Life Cycle (STLC) is a series of particular tasks carried out throughout the testing process to guarantee that software quality objectives are accomplished.

Testing Phases



Test the final System

Test the entire System

Testing the Integrated Component

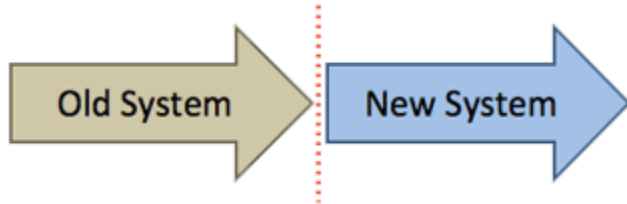
Test Individual Component

6. Deployment

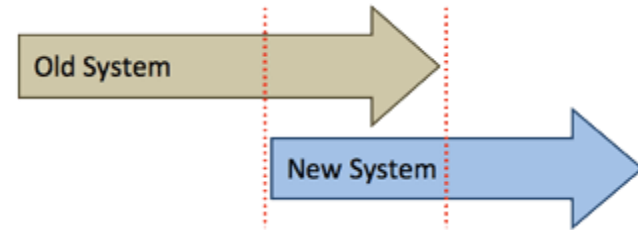


- Deploy the software to the production environment or release it to end-users.
- Ensure a smooth transition from development to the operational environment.

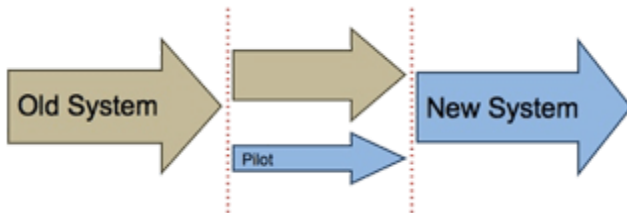
Deployment Phases



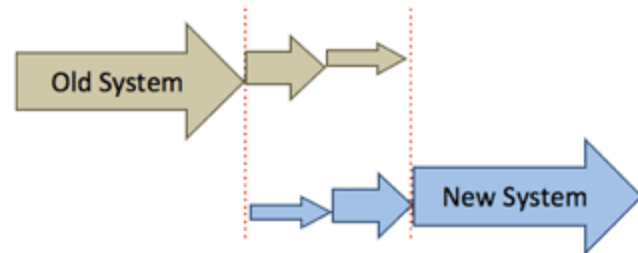
Direct Deployment



Parallel Deployment



Pilot Deployment



Phases Deployment

7. Maintenance and Support

- Address and fix issues reported by users in the production environment.
- Implement updates, enhancements, or patches as needed.

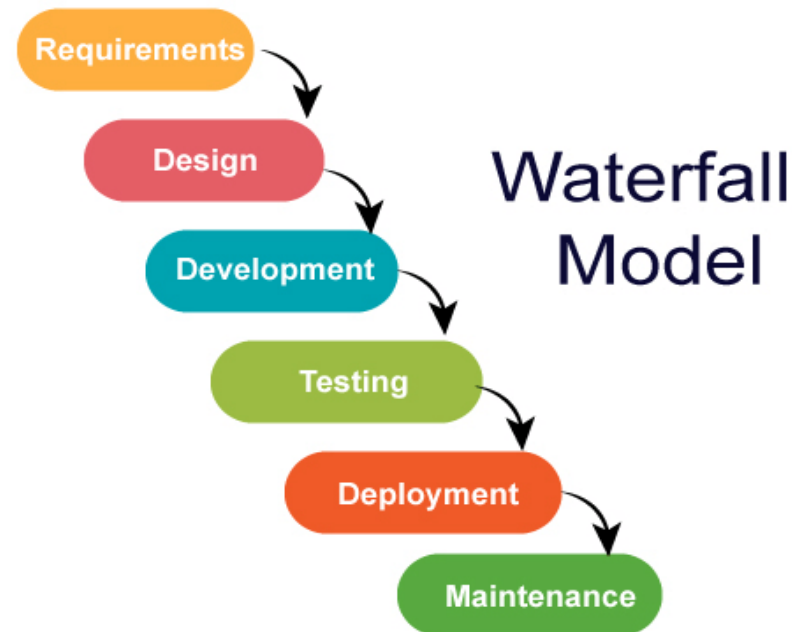
Development Methodologies



- Software Development Life Cycle (SDLC) models are **structured frameworks used to guide the process of developing software applications from conception to deployment and maintenance.**
- These models provide a **systematic approach** to managing the various stages of software development, ensuring that projects are well-organized, predictable, and efficient.

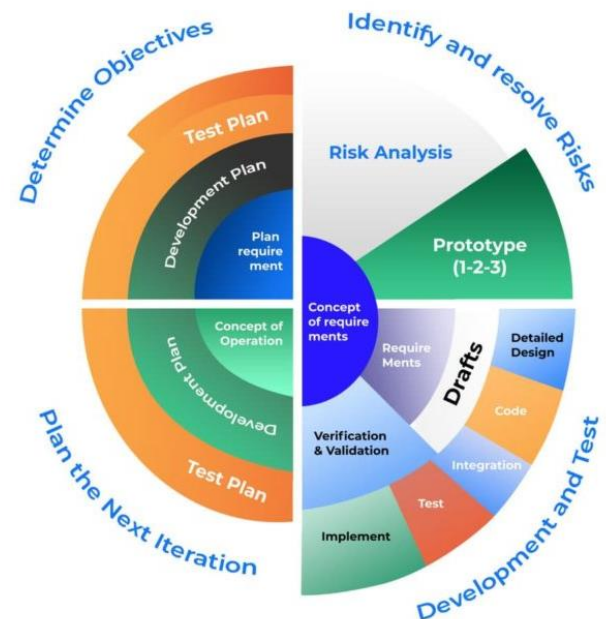
WaterFall Model

- Sequential approach with distinct phases (requirements, design, implementation, testing, deployment, maintenance).
- Progress flows in one direction, with each phase dependent on the completion of the previous one.
- Well-suited for projects with clearly defined and stable requirements.



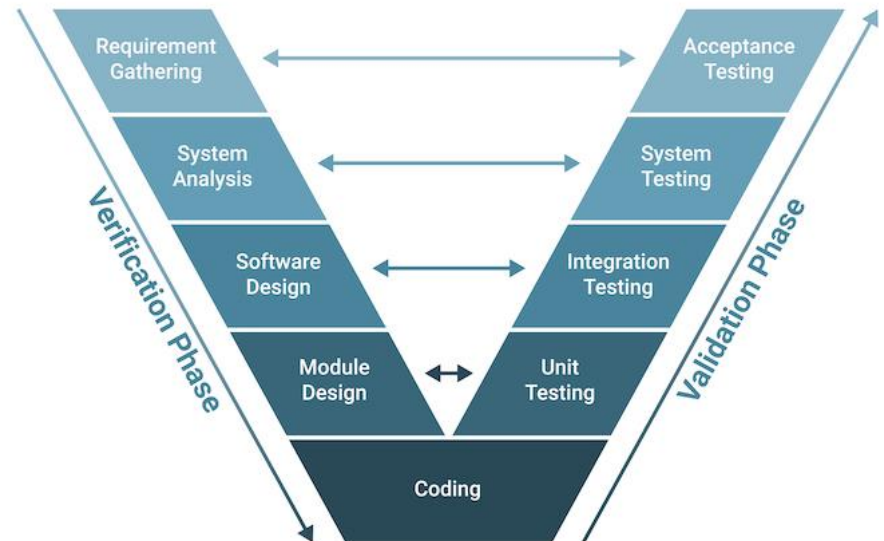
Spiral Model

- Combines elements of both waterfall and iterative models.
- Iteratively cycles through phases of risk analysis, prototyping, development, and testing.
- Each iteration (spiral) focuses on addressing high-risk aspects of the project.
- Well-suited for large, complex projects with evolving requirements and high levels of uncertainty



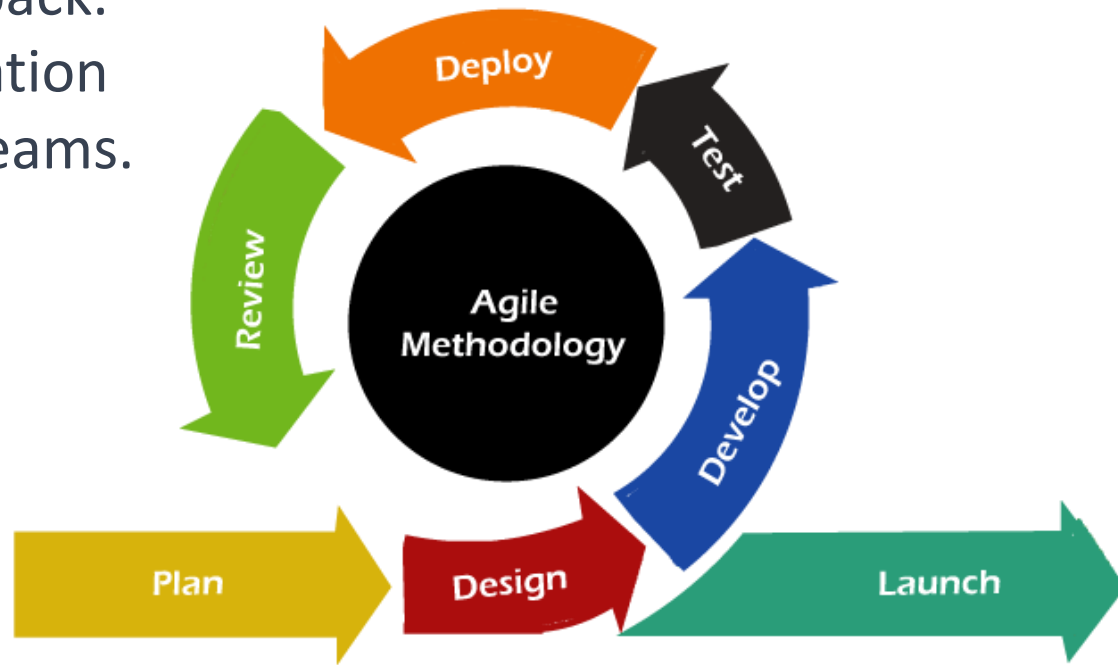
V model (Verification and Validation)

- Extension of the waterfall model that emphasizes the importance of testing and validation.
- Corresponds testing phases (unit testing, integration testing, system testing, acceptance testing) to development phases.
- Ensures that each development phase is accompanied by a corresponding testing phase, leading to better verification and validation of the software.



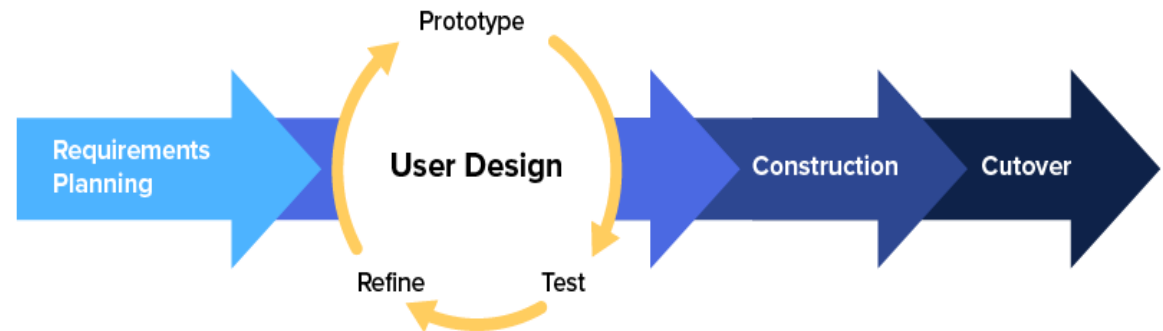
Agile Model

- Emphasizes flexibility, collaboration, and incremental development.
- Iterative approach with short development cycles (sprints), typically lasting 1-4 weeks.
- Prioritizes delivering working software frequently and adapting to customer feedback.
- Encourages close collaboration between cross-functional teams.



RAD Model

- Prioritizes rapid prototyping and iterative development to accelerate the delivery of software projects
- Quick Response to Changing Requirements
- Early Feedback and Validation
- Suitability for Small to Medium-Sized Projects



Home Work



- Do a research on following models
 - Prototyping
 - DevOps Models
 - Agile Model
 - Scrum
 - Kanban
 - Extreme Programming(XP)
 - FDD(Feature Driven Development)
 - Lean Software Development